

# TriVerse AI: Credit Scoring Technical Report

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## Executive Summary

This report evaluates the implementation and optimization of predictive credit scoring classification models on the TriVerse AI monorepo platform. The objective is to determine the probability of a client defaulting on payments over a two-year horizon. Utilizing historical credit files, we compared five machine learning algorithms. The final deployment leverages a CatBoost classifier trained on local GPU hardware (RTX 5060), achieving a peak ROC-AUC of 0.865.

## Dataset Description

The model runs primarily on the Kaggle *Give Me Some Credit* dataset (150,000 instances) and benchmarks against the classic *Statlog German Credit* dataset. Key features include age, debt ratio, monthly income, revolving utilization of unsecured lines, number of open credit lines, and historical late payment tallies.

## Data Cleaning & Preprocessing

Data cleaning handles missing values for columns such as *MonthlyIncome* (imputed via median values grouped by age brackets) and *NumberOfDependents* (imputed to 0). Extremely high debt ratios and utilization scores were capped using 99th percentile winsorization to improve statistical robustness during neural network training.

## Feature Engineering

Preprocessing features includes scaling continuous values with `StandardScaler` and encoding categorical attributes. We engineered interaction terms between revolving utilization and overall debt ratio to capture compound distress indicators, which proved highly predictive in tree-based models.

## Model Training & Optimization

We trained five models: Logistic Regression, Decision Trees, Random Forests, CatBoost, and a Multi-Layer Perceptron (MLP). Hyperparameters were tuned automatically using the Optuna framework via a Bayesian search algorithm over 50 trials, maximizing the ROC-AUC score.

## Results & Comparison

CatBoost outperformed all other models, yielding an ROC-AUC of 0.865 and overall validation accuracy of 93.8%. The Random Forest model achieved 0.851 ROC-AUC, while Logistic Regression yielded 0.803. MLP reached 0.842 ROC-AUC. CatBoost demonstrated optimal speed-to-accuracy ratios on the RTX 5060 GPU.

## Explainability (XAI)

The system integrates SHAP and LIME to generate local risk explanations. The UI renders force plots showing that *RevolvingUtilizationOfUnsecuredLines* is the most dominant factor pushing default risk higher, while higher *age* pushes the score lower.

## Conclusions & Future Work

Deploying the CatBoost model into the FastAPI backend allows for stable, sub-5ms real-time credit checks. Future enhancements will focus on integrating transactional database streaming data to track credit score changes dynamically and training deep neural architectures on larger distributed datasets.